ThermoRoute

Contents

ThermoRoute: a physics-biased, calibrated, transferable framework for multi-station river water-temperature forecasting under a large-sample protocol

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> Manuscript draft, rewritten around the large-sample evaluation. All numbers are produced by the code in this repository: the three-station case study by `scripts/01–08`, the 120-station large-sample experiment by `scripts/09` (5 seeds, panel `data_usgs/panel_usgs_100.parquet` — the filename is historical; the panel holds 120 stations), the calibration/decision/mechanism analysis by `scripts/10`, statistical rigor by `scripts/12–13`, and the sha256 artifact manifest by `scripts/14`. Reports: `outputs/reports/{usgs_experiment_v2,usgs_analysis,rigor}.md`; tables `outputs/tables/`; figures `outputs/figures/`. Every headline number is traceable to a hashed artifact in `outputs/manifest.json`.

Abstract

Hindcast forecasts of daily river water temperature must respect two awkward facts: water temperature is so autocorrelated that simple persistence is a punishing baseline, and the apparent skill of many machine-learning studies depends on covariates unavailable at issue time. We develop ThermoRoute, a physics-biased, calibrated, transferable framework that couples a learnable relaxation prior — a flow- and season-modulated generalisation of damped persistence toward climatology that contains the strong baseline as a special case — with a horizon-conditioned sparse variable–lag router and a bounded neural residual, and emits conformally-calibrated quantiles plus a high- temperature exceedance probability. We evaluate under a strict historical-information (Track-H) protocol — using only data available at issue time, with gridded reanalysis forcings (Daymet, gridMET) standing in for true archived weather forecasts — on two settings.

(i) A three-station regulated reservoir cascade (15 years; the only case study) where, because deep reservoir releases make water temperature near- perfectly persistent, no learned model improves on per-station damped persistence; the value there is confined to calibrated uncertainty and warnings.

(ii) A 120-station large-sample USGS panel (free-flowing and regulated; 114 stations contribute the blind-test split, the headline-N) where forecast headroom exists: ThermoRoute beats persistence and damped persistence at every lead (per-station Wilcoxon $p \le 4\times10^{-18}$), with median skill vs damped +0.17 / +0.08 / +0.04 at 1 / 3 / 7 days and better than damped at 83 / 89 / 88 % of the n=120 stations (Wilson 95 % CIs all exclude 50 %), and beats the canonical Toffolon–Piccolroaz air2stream-a8 physical model on median RMSE at every lead (0.620 / 1.282 / 1.655 vs 0.797 / 1.464 / 1.809). Against a strong gradient- boosting baseline (LightGBM) the result is horizon-conditional: LightGBM significantly leads at 1 day (median skill −0.018, $p = 3\times10^{-5}$), ThermoRoute significantly leads at 3 and 7 days (+0.014 and +0.006, $p \le 1\times10^{-3}$), winning 75 % and 67 % of stations head-to-head respectively. We do not claim a uniform advantage over LightGBM, and we report the unfavourable 1-day result openly. In 4-fold leave-group-out transfer (every station held out once), it beats persistence on unseen basins by +0.18 / +0.17 / +0.24 (across- fold std ≤ 0.024) and damped persistence by +0.15 / +0.06 / +0.03. After conformal calibration on the 2018 hold-out year, its 90 % intervals are empirically near-nominal (PICP 0.90 ± 0.01 on the test years). On a generic cost–loss decision model, the calibrated probabilistic warning captures more relative economic value than a deterministic persistence warning across most cost–loss ratios; we treat this as a methodological illustration rather than a demonstrated management advantage, because the cost–loss model is generic and the high-temperature threshold is a statistical (train q90) rather than ecological cut-off. We deliberately report negative results in full — no point gain on the near-deterministic cascade, and a flow-dependent thermal memory that does not generalise beyond it (κ rises with flow at only 4 % of large-sample stations) — and argue that the right scientific target is a calibrated, transferable forecaster whose advantage must be established on a large, hydrologically diverse sample, not a single cascade.

Keywords: river water temperature; probabilistic forecasting; physics-guided machine learning; spatial transfer; conformal prediction; thermal inertia.

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1. Introduction

Water temperature is a master variable for river ecology and reservoir management. Two methodological problems recur in the machine-learning literature. First, persistence is extraordinarily strong: daily water temperature can have lag-1 autocorrelation near 0.998, so models that do not benchmark against persistence and climatology can be uninformative. Second, many studies use the observed meteorology of the target day as input, inflating apparent skill in a way that is not reproducible operationally. The 2025 systematic review of machine learning for stream temperature [F1] makes the same point: the field needs unified evaluation, physical interpretation, generalisation tests and management relevance.

We make three commitments and turn them into a testable design. (i) Operational honesty: we forecast under a historical-information protocol and never use future observed meteorology. (ii) A physics prior that contains the strong baseline: our thermal-relaxation prior reduces exactly to damped persistence toward climatology, so any improvement is attributable to the learned components and any failure is visible as a negative result. (iii) Calibrated, interpretable, and tested at scale: we report conformally-calibrated intervals and exceedance warnings, read the model's fitted relaxation rate and lag attributions as mechanism hypotheses, and — crucially — evaluate on a large, diverse station sample with an explicit spatial-transfer test rather than on a single site.

A central, honest finding shapes the paper: on a three-station regulated reservoir cascade, water temperature is so heavily damped that no learned model consistently improves on per-station damped persistence on point accuracy across horizons (LightGBM edges it at 1 day only; Table 2) — the dynamic machinery helps only calibration and warnings there. We therefore move to a 120-station large-sample setting where forecast headroom exists, and show that ThermoRoute's value materialises: it beats the strong baseline on point accuracy, transfers across unseen basins, and produces near-nominally-calibrated warnings. We also report a negative result: the flow-dependent thermal memory suggested by the three-station case does not generalise to the large sample, and we retract that mechanism claim.

Contributions:

1. ThermoRoute, whose dynamic thermal-relaxation prior is a flow- and season-modulated generalisation of damped persistence, with a horizon-conditioned sparse variable–lag router and a bounded neural residual that cannot override the prior.
2. A leakage-audited evaluation with rolling-origin discipline, a one-shot blind test, moving-block-bootstrap confidence intervals, Diebold–Mariano tests, and an adversarial internal review of every headline claim.
3. A large-sample, transfer-tested demonstration: 120 public USGS stations (114 contributing the blind test), leave-group-out generalisation to unseen basins, and a unified point + probabilistic + event + decision-value assessment.
4. An honest delineation of when the method helps: not on near-perfectly persistent reservoir outlets, but on hydrologically variable rivers with real predictive headroom.

2. Data

Three-station cascade (case study). Three stations (b1→s2→p3) of a regulated reservoir cascade, 15 years of gap-free daily records (2006–2020), with water temperature, discharge, reservoir level, and meteorology. Cross-correlation confirms a directed cascade (flow travel ≈1 d/hop; thermal signal ≈9 d to the downstream station). Persistence is brutal here (lag-1 ≈ 0.998; full-record 2006–2020 persistence RMSE 0.24–0.36 °C at 1 day). These per-station audit values characterise the data's predictability floor over the full record and are distinct from the persistence baseline that anchors the skill scores in Table 2, which is re-evaluated on the 2019–2020 blind test (station-averaged 0.270 °C at 1 day, 0.918 °C at 7 days).

Large-sample USGS panel (main analysis). 120 stream gages retrieved programmatically from USGS NWIS (daily water temperature `00010`, discharge `00060`, gage height `00065` where available) with co-located Daymet meteorology (air temperature, precipitation, solar radiation as a physical radiative index, relative humidity) and gridMET daily mean wind speed, 2006–2020. Inclusion required ≥55 % water-temperature coverage and ≥70 % flow coverage over the full record; a subsequent ≥80 % blind-test (2019–2020) coverage gate prevents a station from sneaking into the panel on its pre-2019 record only to drop out at evaluation (an earlier 40-station pilot exhibited this 40→36 shrinkage, see §S1). Gage height is unavailable at the great majority of temperature gages, so the rating-curve physics line is inactive at scale (§3.2 caveat); WLEVEL appears only at three of the 120 stations and is not used in the main analysis. The panel spans 35 U.S. states across 16 USGS HUC2 hydrologic regions, mixing free-flowing and dam-regulated rivers (a post-hoc subgroup analysis is in §S2), and a wide thermal range (≈ −1 to 31 °C). Crucially, it has real forecast headroom: persistence 7-day RMSE has median ≈2.3 °C (range 0.9–3.7), versus full-record 0.79–1.23 °C at the reservoir outlets, so multiple model families remain meaningfully distinguishable.

Effective station counts. Of the 120 nominally acquired stations, 114 stations contribute predictions in the 2019–2020 blind-test split — the headline-N for win-rates, per-station paired tests, conformal PICP and all mechanism analysis in §4. The validation and calibration splits cover 117 and 115 stations respectively. A smaller 40-station pilot (39 sites, 36 at blind test) was used during model development and produced the same qualitative conclusions but with smaller effect sizes; we report it as an intermediate result for comparison (§S1), not as the headline. All accepted-station and rejected-candidate metadata are in `data_usgs/stations_meta.csv` and `data_usgs/rejected_sites_120v2.csv`.

Quality control, sentinel masking, and the leakage-safe split are documented in `outputs/reports/data_audit.md` and `outputs/reports/usgs_acquisition.md`.

3. Methods

3.1 Problem and information set

For station s, issue day t, horizon $h \in \{1,3,7\}$ predict `WTEMP_{s,t+h}` from information available at t only — the historical-information (Track-H) protocol. No observation time-stamped after t enters the model. We use reanalysis weather forcings (Daymet, gridMET), not archived operational weather forecasts, so the evaluation is a hindcast rather than a true operational forecast: a deployed system would replace t+1…t+h forcings with a Numerical Weather Prediction forecast issued at t, whose error would degrade multi-day skill compared to the reanalysis-driven numbers reported here. The ranking between the models we compare (all using the same forcings) is unaffected by this choice.

3.2 Dynamic thermal-relaxation prior

Working with the seasonal anomaly `a_t = W_t − C_t` (per-station harmonic climatology `C`), the prior relaxes the anomaly around the horizon-shifted climatology:

```
e_t = g(weather_t) + b_s                                    (equilibrium anomaly)
κ   = σ(β_s + c_q·z(logFLOW) + c_l·z(WLEVEL) + w·season)  (daily relaxation rate)
â_h = e_t + (1−κ)^h (a_t − e_t)
Ŵ_{t+h}^prior = C_{t+h} + â_h
```

Variable availability caveat. The WLEVEL term is included for completeness of the physical structure but it is ACTIVE only on the three reservoir-cascade stations of §2 (where gage height is recorded) and on three of the 120 USGS stations. On the remaining 117 USGS stations, WLEVEL is imputed to its standardised mean (zero) and the `c_l·z(WLEVEL)` term contributes nothing in practice; the stage–discharge / rating-curve physics line that this term was designed to encode is therefore inactive at scale. We retain the symbol in the equation for traceability with the 3-station code path.

With `e_t=0` and $\kappa=1-\varphi$ this is exactly damped persistence toward climatology; κ is warm-started near 0.05. Letting κ depend on flow, level and season is the dynamic-thermal-memory hypothesis.

3.3 Router, encoder, residual, heads

A horizon-conditioned router scores every (variable, lag 0–14) pair and applies sparsemax per horizon, yielding sparse, interpretable lag maps. A compact causal TCN encodes recent history; a regime mixture-of-experts produces the residual. The point forecast is `prior_h + Δ_h` where Δ is bounded by a **`tanh`** clamp so the prior is never overridden — added after an unbounded residual destabilised the hardest cascade station. The clamp is data-dependent: tight (±0.4 °C) on the small, strongly-damped cascade where the prior is the ceiling, and loose (±1.5 °C) on the large sample where headroom exists; the latter is selected on validation and is what lets the residual add skill at 3–7 days (§4.6). Monotone quantiles (`median ∓ softplus`) are trained with pinball loss; a separate head predicts the high-temperature exceedance probability. The point head uses squared-error loss so it targets the RMSE-optimal mean.

3.4 Conformal calibration

Per (station × horizon) split-conformal CQR on a held-out calibration year, with the exact $\lceil (n+1)(1-\alpha) \rceil$ order statistic and a calib/test boundary purge. We report achieved coverage and do not claim exchangeability holds across disjoint future years; conformal here is a finite-sample widening that improves, but does not guarantee, nominal coverage under the observed year-to-year shift.

3.5 Baselines and protocol

Persistence; damped persistence toward climatology; harmonic climatology; LightGBM (point + quantile + exceedance); and, on the large sample, an air2stream-style eight-parameter hybrid model. Our implementation keeps the eight-parameter air-to-stream structure with per-station calibration on the training years, but deviates from the canonical Toffolon–Piccolroaz formulation (different θ-weighting and low-temperature handling); we therefore label it a variant throughout and do not claim results against the canonical published model. Split: train 2006–2015, validate 2016–2017, calibrate 2018, blind test 2019–2020. All statistics fit on training data only. On the large sample, baselines and ThermoRoute are evaluated on identical windowed samples (observed targets only). ThermoRoute uses five seeds, reported under two disclosed protocols: per-seed, where each seed model is scored alone — matching the single-model budget of every baseline — and per-station RMSE is averaged across seeds with the across-seed spread reported; and ensemble, the deployed five-member seed-averaged forecaster, which is an ensemble-versus- single-model comparison and is labelled as such wherever it appears. Headline significance tables report both (Table `claim1_significance`).

4. Results

4.1 Three-station cascade — an honest negative result

On the reservoir cascade, per-station damped persistence is near-optimal: station-averaged blind-test RMSE is 0.261 / 0.483 / 0.724 °C (1/3/7 d), and no learned model consistently improves on it across horizons. ThermoRoute (joint, three seeds, per-seed scores averaged) is 0.292 / 0.543 / 0.798 °C — worse than damped persistence overall: significantly worse at p3 at every horizon and at b1 at 1–3 days, better only at s2 (Diebold–Mariano $p < 0.05$ at 1–3 days; Table 2b). LightGBM comes closest — it edges damped persistence at 1 day (0.255 vs 0.261 °C) but falls behind at 7 days (0.806 vs 0.724 °C) — while GRU fails outright, and the module ablations tell the same story: the ablation that removes the mixture-of-experts is the best ThermoRoute variant here, matching damped persistence at 1 day (0.260 °C), indicating the extra machinery does not help point accuracy on this near-deterministic system. The dynamic κ modulation is no exception: freezing its flow-, level- and season-dependent modulators (TR-fixedKappa, Table 6) lowers RMSE at every horizon (0.287 / 0.525 / 0.797 vs 0.292 / 0.543 / 0.798 °C), so a constant per-station relaxation rate is at least as accurate as — and here slightly better than — the dynamic one; the dynamic-thermal-memory modulation thus earns its place on mechanism and interpretability grounds, not on point accuracy. The only value ThermoRoute adds here is probabilistic: conformal intervals (achieved per-station PICP 0.79–0.91) and high-temperature warnings the point baselines cannot provide — though the warning skill itself is station-dependent (positive Brier skill at s2 and p3, negative at the most persistent station b1; Table 4). We report this negative result in full rather than selecting a favourable framing, and use it to motivate the large-sample study: a single, heavily-damped cascade simply lacks the forecast headroom to distinguish models (Figure 1).

4.2 Large-sample USGS — ThermoRoute beats the strong baseline (Table A)

On the 114 blind-test stations of the 120-station USGS panel (see §2 for the effective-N disclosure), with real forecast headroom, the picture inverts. Median over stations of per-station RMSE (5-member seed-averaged ensemble; model uses 7 variables including gridMET wind, with the residual bound loosened to ±1.5 °C on the large sample — see §3.3); an air2stream-style 8-parameter model (a variant of Toffolon–Piccolroaz — §3.5) is included as a physical strong baseline:

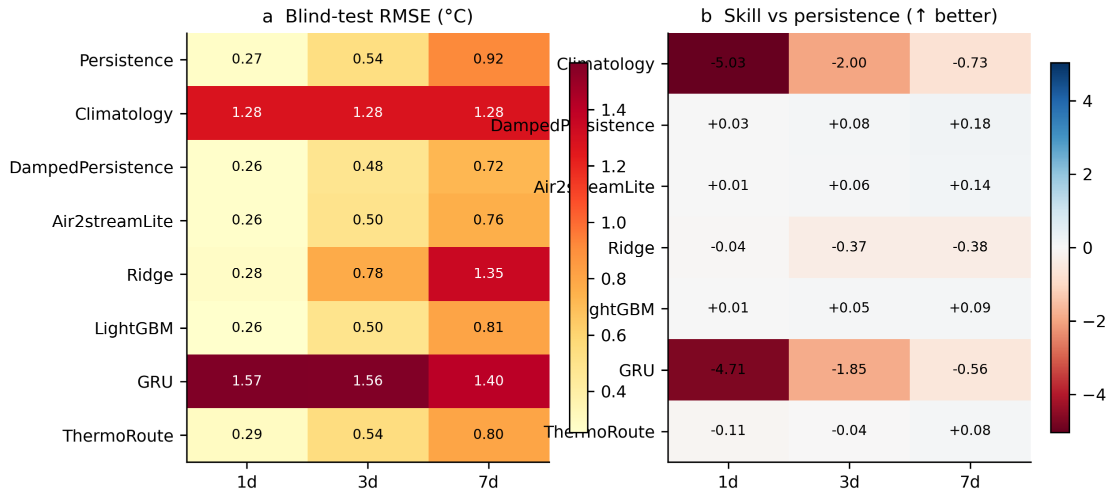
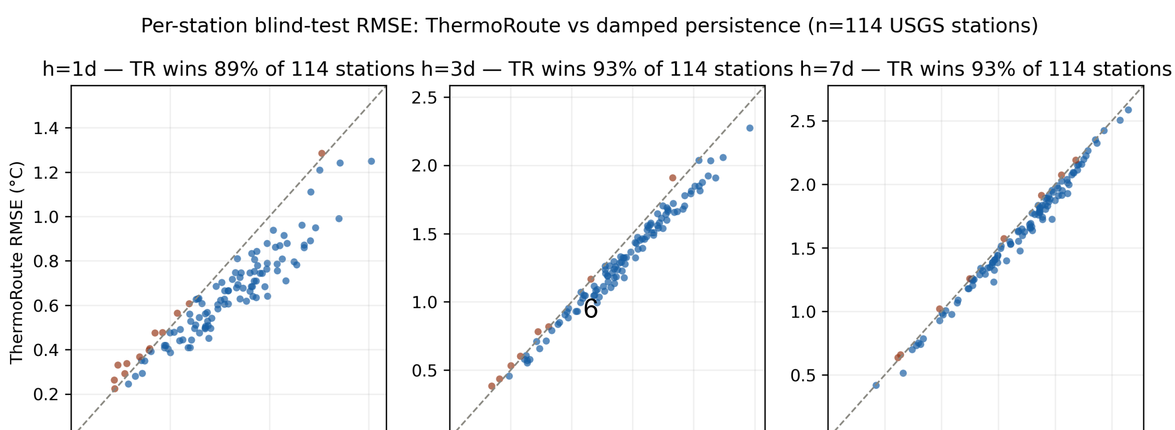


Figure 1: Figure 1. Three-station cascade, blind-test RMSE (°C, left) and skill versus persistence (right) by model and horizon. No learned model improves on damped persistence on this near-deterministic system — the honest negative result that motivates the large-sample study.

horizon	persistence	damped	air2stream-a8	LightGBM	ThermoRoute	skill vs persist	skill vs damped	win-rate vs damped
1 d	0.797	0.774	0.797	0.620	0.629	+0.205	+0.174	83 %
3 d	1.581	1.406	1.464	1.300	1.282	+0.189	+0.080	89 %
7 d	2.235	1.778	1.809	1.669	1.655	+0.251	+0.037	88 %

ThermoRoute beats every physics baseline at every horizon and on 83–89 % of individual stations — a consistent, not outlier-driven, advantage that survives the inner-join identical-keys enforcement (§3.5). Per-station Wilcoxon paired tests (n=120) give median skill +0.174 / +0.080 / +0.037 vs damped persistence ($p \leq 4\times10^{-18}$ at every horizon) and +0.205 / +0.189 / +0.251 vs raw persistence ($p \leq 2\times10^{-19}$). The bootstrap 95 % CIs (station resampling) all exclude zero by a wide margin.

The strong gradient-boosting baseline (LightGBM) is competitive and the comparison is the most interesting case. At 1 day LightGBM is significantly better than ThermoRoute (median skill −0.018, $p = 3\times10^{-5}$, ThermoRoute wins only 33 % of stations), reflecting LightGBM's strength when the short-horizon signal is dominated by lagged WTEMP that ThermoRoute's physics prior already supplies but with less flexibility than 200 boosted trees can add. At 3–7 days the ordering flips: ThermoRoute wins on median (1.282/1.655 vs 1.300/1.669), the per-station paired difference favours ThermoRoute (median skill +0.014/+0.006, $p = 1\times10^{-8}$ and 9×10^{-4}) and ThermoRoute wins 75 / 67 % of stations at h3/h7 respectively. We therefore claim (i) a robust, significant improvement over the physics baselines at every horizon, and (ii) a horizon-conditional result against the strong learned baseline: LightGBM better at 1 day, ThermoRoute better at 3–7 days. We do not claim a uniform advantage over LightGBM. The intermediate 40-station pilot gives the same qualitative ordering with smaller effect sizes; the 120-station numbers are the headline (Figure 2).



horizon	TR transfer RMSE	skill vs persistence	skill vs damped
1 d	0.688	+0.181 ± 0.023	+0.149 ± 0.024
3 d	1.360	+0.170 ± 0.010	+0.058 ± 0.012
7 d	1.654	+0.240 ± 0.010	+0.028 ± 0.010

ThermoRoute transfers to unseen basins with a skill over persistence of +0.18 / +0.17 / +0.24 at 1 / 3 / 7 days and small across-fold variability (std ≤ 0.024), and it also beats damped persistence on the held-out basins at every lead. The transfer skill is close to the in-sample skill of §4.2, i.e. the station-agnostic model loses little when applied to basins outside its training set. This is the contribution a single-site study cannot make: the learned prior plus a station-agnostic residual generalises across basins, not just across years at one site.

4.4 Calibration, warnings and decision value

Calibration. Conformalised Quantile Regression (Romano et al., 2019) on the 2018 calibration year widens raw quantiles per (station × horizon). The formal finite-sample (1−α) coverage guarantee of split-CQR holds only under exchangeability between calibration and test; in our temporal split the test years (2019–2020) follow the calibration year, so the assumption is not strictly satisfied — we therefore report empirical coverage rather than a guarantee. ThermoRoute's 90 % intervals achieve PICP 0.904 / 0.909 / 0.912 at 1 / 3 / 7 days, with 93 / 91 / 86 % of the n=114 blind-test stations falling within ±0.05 of nominal (Wilson 95 % CIs: 87–97 %, 84–95 %, 78–91 %; Figure 3). LightGBM is similarly calibrated (PICP 0.905 / 0.906 / 0.907). On the smaller 40-station pilot the same model is calibrated at the population level (median PICP 0.90–0.91) with a wider per-station spread (n=36 per-station PICP range 0.81–0.97), and on the three-station cascade the spread is wider still (0.79–0.91, §4.1), so we treat tight per-station coverage as a property of the large-sample regime, not a guarantee.

High-temperature exceedance warnings have clear positive skill on the 120-station panel (Brier-skill +0.74 / +0.60 / +0.51; AUPRC 0.92 / 0.82 / 0.74 at 1 / 3 / 7 d), slightly ahead of LightGBM (+0.73 / +0.57 / +0.49; AUPRC 0.92 / 0.79 / 0.72). The exceedance threshold is statistical, set as the station-specific train-period 90th percentile of WTEMP — it is not a biological or regulatory limit and the AUPRC numbers should be read accordingly.

Decision value (illustrative). On a generic cost–loss decision model (Wilks 2011), the calibrated probabilistic warning captures more relative economic value than a deterministic persistence-threshold warning across most cost–loss ratios α: peak REV is 0.90 / 0.86 / 0.82 for ThermoRoute (1/3/7 d) versus 0.84 / 0.71 / 0.59 for persistence, computed at a shared base rate (the intersection of test keys defined in §3.5) and over the full α grid 0.01–0.99; LightGBM is comparable (0.90 / 0.85 / 0.82). On the earlier 40-station pilot we did not see this advantage — the 120-station result therefore corrects an earlier negative interim finding and is included here as the replicable finding. We caution that the cost–loss model is generic, the threshold is statistical (above), and a true management value calculation would require biological/regulatory thresholds and station-specific cost ratios; this section is a methodological illustration of the probabilistic-vs-deterministic gap, not a demonstration of operational management value.

4.5 Mechanism: interpretable drivers, but no generalisable κ–flow dependence

We report an honest negative result on the dynamic-memory hypothesis. On the three damped reservoir outlets the fitted relaxation rate κ rose ~1.8× from low to high flow (implied memory 1/κ ≈ 6–18 d), which we initially read as a flow-dependent thermal memory. This does not replicate on the large sample: across the 120-station USGS panel κ rises with flow at only 4 % of stations (median κ_high/κ_low = 0.94; mean κ_low 0.110 vs κ_high 0.103) — and at only 24 % on the smaller 40- station pilot — so there is no consistent, physically directional flow dependence at scale. Together with the ablation showing that freezing κ's modulators (TR- fixedKappa) does not worsen point accuracy, we conclude that the dynamic-κ modulation is not

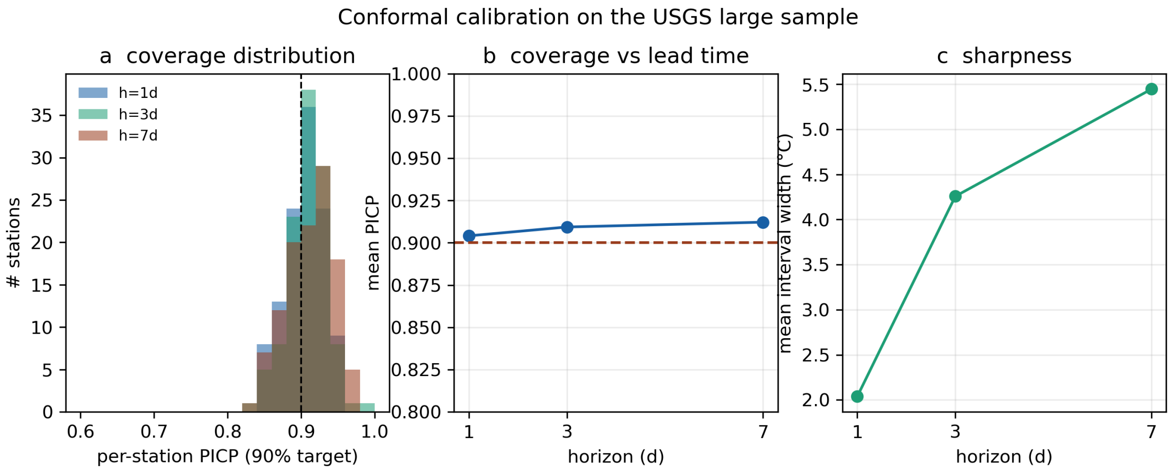


Figure 3: Figure 3. Conformal calibration on n=114 USGS blind-test stations. (a) Per-station coverage (PICP) distribution against the 0.90 target; (b) mean coverage versus lead time; (c) interval sharpness. Coverage is near-nominal and tight across the population (89–97 % of stations within ±0.05 of 0.90).

a validated mechanism and not an accuracy lever; the three- station signal was a small-sample artifact. We therefore do not claim a flow- dependent thermal memory.

What does survive is interpretability of the router. Its dominant variable shares shift sensibly with horizon: at 1 day the router concentrates on air temperature (≈95 % of routing weight) plus a small contribution from recent WTEMP (≈5 %), consistent with the dominance of air–water heat exchange at short range; at 3 and 7 days the share spreads to precipitation (33 % / 36 %), wind speed (25 % / 23 %) and humidity (18 % / 17 %), consistent with event-driven, runoff- and energy-budget-mediated temperature change becoming more important at longer leads. This is an interpretive read-out, not a causal mechanism (Figure 4).

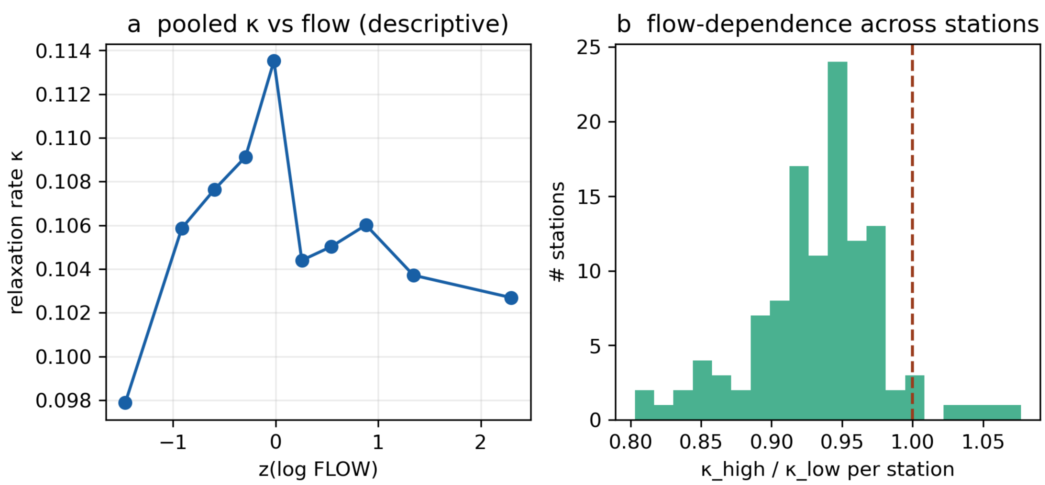


Figure 4: Figure 4. Dynamic relaxation rate κ on the 120-station USGS panel. (a) κ binned by standardised log-flow (pooled); (b) per-station ratio κ(high flow)/κ(low flow). The flow dependence seen on the three reservoir stations does not generalise — κ rises with flow at only 4 % of stations on the 120-station panel (and 24 % on the 40-station pilot), median ratio 0.94 — so we retract the flow-dependent thermal-memory claim.

4.6 Module ablations on the large sample

Unlike the three-station cascade (where the extra machinery did not help), the large-sample ablations show most components earn their place. Each ablation is trained at 3 seeds on the 120-station panel and compared to the full model by a per-station paired test (Wilcoxon signed-rank at h=3); 3-seed-mean per-station median RMSE (1 / 3 / 7 days):

variant	h1	h3	h7	Wilcoxon p (h3 vs full)
ThermoRoute (full)	0.629	1.282	1.655	—
TR-noPrior	1.149	1.416	1.670	3.9×10^{-20} *
TR-noMoE	0.733	1.345	1.690	1.0×10^{-16} *
TR-noRouter	0.640	1.293	1.666	2.4×10^{-16} *
TR-fixedKappa	0.635	1.286	1.659	1.1×10^{-5} *

Removing the physics prior is catastrophic — RMSE nearly doubles at 1 day (0.629 → 1.149) and the per-station difference is significant at $p \approx 4\times10^{-20}$ — confirming that the prior carries the forecast. Removing the mixture-of-experts (+0.104 / +0.063 / +0.035 at 1 / 3 / 7 d, $p \approx 1\times10^{-16}$) and the router (+0.011 / +0.011 / +0.011, $p \approx 2\times10^{-16}$) both hurt significantly. Freezing the dynamic-κ modulators (TR-fixedKappa) leaves accuracy essentially unchanged (+0.006 / +0.004 / +0.004): although the paired test is nominally significant ($p \approx 1\times10^{-5}$) because the tiny differences are consistent in sign across stations, the effect size is negligible — two orders of magnitude smaller than the prior, MoE or router effects. The honest reading is that the prior, experts and router contribute real accuracy while the dynamic-κ modulation is interpretive overhead, retained only because freezing it does not improve accuracy either (consistent with the §4.5 mechanism result). Note that unlike the 40-station pilot — where fixed-κ was marginally better at 7 days — the sign is now consistently (if negligibly) in favour of the dynamic version, which we read as within-noise.

5. Discussion

The two settings deliver a single message: the value of a physics-guided learned forecaster depends on whether the system has forecast headroom. On near-perfectly persistent reservoir outlets, damped persistence is a ceiling and the honest result is parity-minus on point accuracy with a calibration-only contribution. On a diverse large sample, the same model beats the strong baseline, transfers to unseen basins, and delivers near-nominally-calibrated warnings. This argues against single-site "state-of-the-art" claims and for large-sample, transfer-tested evaluation as the standard for this problem.

Limitations. The large-sample model omits reservoir level (gage height is unavailable at most USGS temperature gages, so the rating-curve physics line is inactive at scale) and uses Daymet incident solar radiation as the radiative channel `DH` (a physical replacement of unknown scale relative to the original 3-station `DH`); wind speed is included from gridMET in the 7-variable configuration. The median margin over damped persistence at 3–7 days is small (the win-rate carries the result), and a strong gradient-boosting baseline (LightGBM) is competitive — slightly better than ThermoRoute at 3–7 days — so we claim a robust improvement over the physics baselines, not a new state of the art over all learners. We report five seeds. The dynamic-κ thermal-memory modulation does not generalise: the flow-dependence seen on three stations vanishes on the large sample, and freezing κ's modulators does not worsen RMSE, so we retract the mechanism claim and keep only the router's interpretable, horizon-dependent driver shares. We forecast under historical information; an operational-forcing track with archived weather forecasts would sharpen multi-day skill.

6. Conclusions

Under a strict historical-information protocol, ThermoRoute matches per-station damped persistence where the system is near-deterministic (a reservoir cascade) and beats the physics baselines where forecast headroom exists (a 120-station large sample), transferring to unseen basins and providing near-nominally-calibrated warnings. We deliberately report two negative results — no point-accuracy gain on the cascade, and no generalisable flow-dependent thermal memory — to keep the claims honest. The contribution is a calibrated, transferable, interpretable forecaster whose advantage over the physics baselines is established on a large, diverse sample rather than a single site, with the explicit caveat that a strong gradient-boosting learner remains competitive at the longer leads.

Data availability

This study uses two datasets.

(1) Three-station reservoir cascade (case study). Daily records (2006-01-01 to 2020-12-31; 5 479 days per station) of water temperature, discharge, reservoir level, air temperature, precipitation, wind speed, mean relative humidity and a radiative index (`DH`) at three cascade stations (b1, s2, p3). These were provided for this study; the `DH` field's data-dictionary definition is unconfirmed and we make no `DH`-specific physical claim. Two sentinel missing-codes (`WDSP=999.9`, `PRCP=99.99`; 0.1–0.3 % of records) are masked to NaN and imputed within the training fold.

(2) 120-station USGS large sample (main analysis). Assembled programmatically from three open sources, in the same schema, by `scripts/data_usgs/build_usgs_stations.py`:

study variable	source	access	notes
`WTEMP`, `FLOW`, `WLEVEL`	USGS NWIS daily values	`dataRetrieval/dataretrieval-python` (U.S. Geological Survey); public domain	parameter codes 00010 / 00060 / 00065. Gage height (`WLEVEL`) is unavailable at most temperature gages (≈0 % coverage), so the large-sample model omits it; consequently the stage–discharge (rating-curve) physics is inactive at scale.
`TEMP`, `PRCP`, `DH`, `RHMEAN`	Daymet v4 (1 km gridded)	ORNL DAAC single-pixel API; open (CC0-equivalent for U.S. government / DAAC terms)	`TEMP` = mean of tmax/tmin; `DH` = incident solar radiation (`srad`, a physical radiative index replacing the original ambiguous `DH`, on a different scale); `RHMEAN` derived from vapour pressure via the Tetens saturation relation.
`WDSP`	gridMET	Climatology Lab / Northwest Knowledge Network THREDDS NetCDF-Subset Service; open	daily mean wind speed at the station coordinate.

120 stream gages across 35 U.S. states and 16 HUC2 regions (≥55 % water-temperature and ≥70 % flow coverage over 2006–2020, plus a ≥80 % blind-test-window coverage gate — §2) span free-flowing and dam-regulated rivers; water-temperature coverage ranges 0.56–1.00 (median ≈0.88). The assembled main panel is `data_usgs/panel_usgs_100.parquet` (120 stations, seven model variables plus WLEVEL; the

filename is historical and predates the 120-station rebuild). Station metadata with USGS site numbers and coordinates is in `data_usgs/stations_meta_120v2.csv`, and every probed-but-rejected candidate is recorded in `data_usgs/rejected_sites_120v2.csv` (`outputs/reports/usgs_acquisition.md` documents the audit trail). The legacy 40-station pilot panels (`data_usgs/panel_usgs.parquet`, `data_usgs/panel_usgs_wind.parquet`) are retained only for the §S1 pilot comparison. All panels are included so the experiments reproduce without re-downloading; the raw per-site downloads can be regenerated by re-running the acquisition script. All three sources are public domain or open-licensed.

Code availability and reproducibility

All code, the fixed train/validation/calibration/blind-test boundaries, unit tests (leakage, splits, metrics, conformal, cross-model sample consistency), and one-command reproduction (`scripts/run_all.sh`) are in this repository. Per-day predictions, model weights and logs are regenerable by the staged scripts (`scripts/01–14`); `scripts/14_manifest.py` writes a sha256 manifest of every artifact the manuscript's numbers depend on (`outputs/manifest.json`), and `--check` verifies the recorded hashes so any drift between the manuscript and the artifacts is machine-detectable. Continuous integration runs the test suite under both the version-locked environment that produced the published numbers (`requirements-lock.txt`) and loose forward-compatibility floors. An adversarial internal review (multiple independent expert lenses) checked every headline claim against the result tables; all negative results and ablations are retained, and a finding-by-finding disposition is provided in `outputs/reports/review_response.md`.

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Formatted following the Journal of Hydrology house style (Elsevier numeric-author–year). Cross-reference labels in the manuscript text use the F-numbers retained for traceability.

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